

Chapter 3, Section 5

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1. Let X be a projective scheme over a field k , and let $\mathcal{F}$ be a coherent sheaf on X . We define the *Euler characteristic* of $\mathcal{F}$ by

$$\chi(\mathcal{F}) = \sum (-1)^i \dim_k H^i(X, \mathcal{F}).$$

If

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

is a short exact sequence of coherent sheaves on X , show that $\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'')$.

Proof. We get an exact sequence of finite dimensional k -vector spaces (5.2)

$$\cdots \xrightarrow{\delta_{i-1}} H^i(X, \mathcal{F}') \longrightarrow H^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F}'') \xrightarrow{\delta_i} \cdots,$$

where the δ_i 's are connecting homomorphisms. By exactness, we have

$$\begin{aligned} \chi(\mathcal{F}) &= \sum (-1)^i \dim_k H^i(X, \mathcal{F}) \\ &= \sum (-1)^i (\dim_k \operatorname{coker} \delta_{i-1} + \dim_k \ker \delta_i) \\ &= \sum (-1)^i (\dim_k H^i(X, \mathcal{F}') - \dim_k \operatorname{im} \delta_{i-1} + \dim_k H^i(X, \mathcal{F}'') - \dim_k \operatorname{im} \delta_i) \\ &= \sum (-1)^i \dim_k H^i(X, \mathcal{F}') + \sum (-1)^i \dim_k H^i(X, \mathcal{F}'') \\ &= \chi(\mathcal{F}') + \chi(\mathcal{F}''). \end{aligned}$$

□

2. (a) Let X be a projective scheme over a field k , let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X over k , and let $\mathcal{F}$ be a coherent sheaf on X . Show that there is a polynomial $P(z) \in \mathbb{Q}[z]$, such that $\chi(\mathcal{F}(n)) = P(n)$ for all $n \in \mathbb{Z}$. We call P the *Hilbert polynomial* of $\mathcal{F}$ with respect to the sheaf $\mathcal{O}_X(1)$.
- (b) Now let $X = \mathbb{P}_k^r$, and let $M = \Gamma_*(\mathcal{F})$, considered as a graded $S = k[x_0, \dots, x_r]$ -module. Use (5.2) to show that the Hilbert polynomial of $\mathcal{F}$ just defined is the same as the Hilbert polynomial of M defined in (I, §7).

Proof.

- (a) Since $\mathcal{O}_X(1)$ is a very ample sheaf on X over k , there is a closed immersion $i : X \rightarrow \mathbb{P}_k^r$ of schemes over k , for some r , such that $\mathcal{O}_X(1) = i^* \mathcal{O}_{\mathbb{P}^r}(1)$. If $\mathcal{F}$ is coherent on X , then $i_* \mathcal{F}$ is coherent on $\mathbb{P}_k^r$ (II, Ex. 5.5), and the cohomology is the same (2.10). The Euler characteristic is defined in terms of cohomology, so we reduce to the case $X = \mathbb{P}_k^r$.

Cover X by a finite number of open affines $U_i \cong \operatorname{Spec} k[y_1, \dots, y_r]$. Then there exist finitely generated $k[y_1, \dots, y_r]$ -modules M_i such that $\mathcal{F}|_{U_i} \cong \tilde{M}_i$. Since $\operatorname{Spec} A_i \cap \operatorname{Supp} \mathcal{F}$ is a closed subset of $\operatorname{Spec} A_i$; in fact, $\operatorname{Spec} A_i \cap \operatorname{Supp} \mathcal{F} = V(\operatorname{Ann} M_i)$ (AM, Ex. 3.19), $\operatorname{Supp} \mathcal{F}$ is a closed subset of X . We proceed by induction on $\dim \operatorname{Supp} \mathcal{F}$. If $\operatorname{Supp} \mathcal{F}$ has dimension zero, then $\operatorname{Supp} \mathcal{F} \cap U_i$ is affine (II, Ex. 3.11b) equal to spectra of a Noetherian ring A of dimension zero, which are precisely Artinian rings (AM, 8.5). So $U_i \cap \operatorname{Supp} \mathcal{F}$ is a finite set (AM, Ex. 8.2), and since X can be covered by a finite number of U_i , $\operatorname{Supp} \mathcal{F}$ is a finite set of closed points. Thus, $\mathcal{F}$ is isomorphic to a direct sum of skyscraper sheaves. More precisely, $\mathcal{F} = \bigoplus_{P \in \operatorname{Supp} \mathcal{F}} \mathcal{F}_P$, where $\mathcal{F}_P$ is the stalk of $\mathcal{F}$ at P , and cohomology commutes with direct sums, so by (Ex. 1), we reduce to the case when $\operatorname{Supp} \mathcal{F}$ consists of a single point. Then, $\chi(\mathcal{F}(n))$ is just the Hilbert polynomial of $\mathcal{F}_P$ (I, 7.5) as a graded $A[y_1, \dots, y_r]$ -module.

Assume the statement to be true for all coherent sheaves with suPort dimension less than n . Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{x_r} \mathcal{O}_X \longrightarrow \mathcal{O}_H \longrightarrow 0,$$

where H is the hyperplane $x_r = 0$. Taking the tensor product of this sequence with $\mathcal{F}$ gives us

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{F}(-1) \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}_H \longrightarrow 0,$$

where $\mathcal{R}$ is the kernel of the map $\mathcal{F}(-1) \rightarrow \mathcal{F}$. Twisting by $\mathcal{O}_X(n)$ is exact, so we get

$$\chi(\mathcal{F}(n)) - \chi(\mathcal{F}(n-1)) = \chi(\mathcal{F}_H(n)) - \chi(\mathcal{R}(n))$$

by (III, Ex. 5.1). Consider the coherent $\mathcal{O}_X$ -module $\mathcal{R}$. It vanishes under the multiplication map x_r , where x_r locally defines one of the coordinates such that $x_r = 0$ is the defining equation for H , and such that there is a inclusion $\text{Supp } \mathcal{R} \subseteq H$, which exists by Bertini's theorem. Since $H \cong \mathbb{P}_k^{r-1}$, we can apply the induction hypothesis to $\mathcal{R}$ and $\mathcal{F}_H$, so let $P_{\mathcal{R}}(z), P_{\mathcal{F}_H}(z) \in \mathbb{Q}[z]$ be polynomials such that

$$\chi(\mathcal{R}(n)) = P_{\mathcal{R}}(n), \quad \chi(\mathcal{F}_H(n)) = P_{\mathcal{F}_H}(n).$$

The difference function of $\chi(\mathcal{F}(n))$ is equal to $P_{\mathcal{F}_H}(z) - P_{\mathcal{R}}(z) \in \mathbb{Q}[z]$, so by (§1, 7.3), we conclude that $\chi(\mathcal{F}(n)) = P(n)$ for some $P(z) \in \mathbb{Q}[z]$.

- (b) By (5.2), there exists an integer n_0 such that for each $i > 0$ and each $n \geq n_0$, $H^i(X, \mathcal{F}(n)) = 0$. Let $P(n)$ be the Hilbert polynomial just defined, and let $P'(n)$ be the polynomial from associated to the graded S -module M from (I, §7). Then $P(n) = \dim_k H^0(X, \mathcal{F}(n))$ for each $n \geq n_0$, but $M_n = \Gamma(X, \mathcal{F}(n)) = H^0(X, \mathcal{F}(n))$ and $P'(n) = \dim_k M_n$; in particular, $P(n)$ and $P'(n)$ agree at infinitely many values. Hence, they must be equal.

□

- 3. Arithmetic Genus.** Let X be a projective scheme of dimension r over a field k . We define the *arithmetic genus* p_a of X by

$$p_a(X) = (-1)^r (\chi(\mathcal{O}_X) - 1).$$

Note that it depends only on X , not on any projective embedding.

- (a) If X is integral, and k algebraically closed, show that $H^0(X, \mathcal{O}_X) \cong k$, so that

$$p_a(X) = \sum_{i=0}^{r-1} (-1)^i \dim_k H^{r-i}(X, \mathcal{O}_X).$$

In particular, if X is a curve, we have

$$p_a(X) = \dim_k H^1(X, \mathcal{O}_X).$$

- (b) If X is a closed subvariety of $\mathbb{P}_k^r$, show that this $p_a(X)$ coincides with the one defined in (I, Ex. 7.2), which aParently depended on the projective embedding.
- (c) If X is a nonsingular projective curve over an algebraically closed field k , show that $p_a(X)$ is in fact a *birational* invariant. Conclude that a nonsingular plane curve of degree $d \geq 3$ is not rational.

Proof.

- (a) By (II, Ex. 2.4), there exists a natural bijection

$$\text{Hom}_{\text{Sch}(k)}(X, \mathbb{A}^1) \cong \text{Hom}_{\text{Alg}(k)}(k[t], \Gamma(X, \mathcal{O}_X)).$$

The space on the right is naturally isomorphic to $H^0(X, \mathcal{O}_X)$ as vector spaces over k , so it is enough to show any morphism $f : X \rightarrow \mathbb{A}^1$ over k must be constant. Since X is projective over k , and $\mathbb{A}^1$ is a separated scheme over k , by (II, Ex. 4.8) f is a proper morphism. In particular, f is a closed map, so the image of X in $\mathbb{A}^1$ is a closed subscheme. The proper closed subschemes of $\mathbb{A}^1$ are finite sets of points. But the image of a proper scheme is proper (II, Ex. 4.4), and $\mathbb{A}^1$ is not a proper scheme over k . Hence, f must be constant.

- (b) Let $\mathcal{I}_X$ be the ideal sheaf of X on $\mathbb{P}_k^r$. The homogenous coordinate ring $S(X)$ of X for the given embedding is $k[x_0, \dots, x_r]/I$, where I is the ideal $\Gamma_*(\mathcal{I}_X)$ (II, Ex. 5.14). The result follows by (Ex. 1) and (Ex. 2b).

- (c) Two nonsingular projective curves are birational if and only if they are isomorphic (II, 6.7), so they must have the same cohomology. But by the degree-genus formula, any curve of degree ≥ 3 has non-zero arithmetic genus, while $\mathbb{P}^1$ has genus zero.

□

6. Curves on a Nonsingular Quadric Surface. Let Q be the nonsingular quadric surface $xy = zw$ in $X = \mathbb{P}_k^3$ over a field k . We will consider locally principal closed subschemes Y of Q . These correspond to Cartier divisors on Q by (II, 6.17.1). On the other hand, we know that $\text{Pic } Q \cong \mathbb{Z} \oplus \mathbb{Z}$, so we can talk about the *type* (a, b) of Y (II, 6.16) and (II, 6.6.1). Let us denote the invertible sheaf $\mathcal{L}(Y)$ by $\mathcal{O}_Q(a, b)$. Thus, for any $n \in \mathbb{Z}$, $\mathcal{O}_Q(n) = \mathcal{O}_Q(n, n)$.

(a) Use the special cases $(q, 0)$ and $(0, q)$, with $q > 0$, when Y is a disjoint union of q lines $\mathbb{P}^1$ in Q , to show:

- (1) if $|a - b| \leq 1$, then $H^1(Q, \mathcal{O}_Q(a, b)) = 0$;
- (2) if $a, b < 0$, then $H^1(Q, \mathcal{O}_Q(a, b)) = 0$;
- (3) if $a \leq -2$, then $H^1(Q, \mathcal{O}_Q(a, 0)) \neq 0$.

(b) Now use these results to show:

- (1) if Y is a locally principal closed subscheme of type (a, b) with $a, b > 0$, the Y is connected;
- (2) now assume k is algebraically closed. Then for any $a, b > 0$, there exists an irreducible nonsingular curve Y of type (a, b) . Use (II, 7.6.2) and (II, 8.18).
- (3) an irreducible nonsingular curve Y of type (a, b) , $a, b > 0$ on Q is projectively normal (II, Ex. 5.14) if and only if $|a - b| \leq 1$. In particular, this gives lots of examples of nonsingular, but not projectively normal curves in $\mathbb{P}^3$. The simplest is the one of type $(1, 3)$, which is just the rational quartic curve (I, Ex. 3.18).

(c) If Y is a locally principal subscheme of type (a, b) in Q , show that $p_a(Y) = ab - a - b + 1 = (a - 1)(b - 1)$.

Proof.

- (a)
- (b)
- (c)

□

7. Let X (respectively, Y) be proper schemes over a Noetherian ring A . We note by $\mathcal{L}$ an invertible sheaf.

- (a) If $\mathcal{L}$ is ample on X , and Y is any closed subscheme of X , then $i^*\mathcal{L}$ is ample on Y , where $i : Y \rightarrow X$ is the inclusion.
- (b) $\mathcal{L}$ is ample on X if and only if $\mathcal{L}_{\text{red}} = \mathcal{L} \otimes \mathcal{O}_{X_{\text{red}}}$ is ample on X_{red} .
- (c) Suppose X is reduced. Then $\mathcal{L}$ is ample on X if and only if $\mathcal{L} \otimes \mathcal{O}_{X_i}$ is ample on X_i , for each irreducible component X_i of X .
- (d) Let $f : X \rightarrow Y$ be a finite surjective morphism, and let $\mathcal{L}$ be an invertible sheaf on Y . Then $\mathcal{L}$ is ample on Y if and only if $f^*\mathcal{L}$ is ample on X .

Proof.

- (a)
- (b)
- (c)
- (d)

□

10. Let X be a projective scheme over a Noetherian ring A , and let $\mathcal{F}^1 \rightarrow \mathcal{F}^2 \rightarrow \cdots \rightarrow \mathcal{F}^r$ be an exact sequence of coherent sheaves on X . Show that there is an integer n_0 , such that for all $n \geq n_0$, the sequence of global sections

$$\Gamma(X, \mathcal{F}^1(n)) \longrightarrow \Gamma(X, \mathcal{F}^2(n)) \longrightarrow \cdots \longrightarrow \Gamma(X, \mathcal{F}^r(n))$$

is exact.

Proof. Denote $f_i : \mathcal{F}^i \rightarrow \mathcal{F}^{i+1}$ for each $i = 1, \dots, n-1$. For every $i = 2, \dots, n-1$, we have the following three short exact sequence

$$0 \longrightarrow \operatorname{im} f_{i-1} \longrightarrow \mathcal{F}^i \longrightarrow \operatorname{im} f_i \longrightarrow 0,$$

$$0 \longrightarrow \ker f_{i-1} \longrightarrow \mathcal{F}^{i-1} \longrightarrow \operatorname{im} f_{i-1} \longrightarrow 0,$$

$$0 \longrightarrow \operatorname{im} f_i \longrightarrow \mathcal{F}^{i+1} \longrightarrow \operatorname{coker} f_i \longrightarrow 0.$$

Every sheaf above is coherent (II, 5.7). By (5.2), there exists an integer n_i such that for each $i > 0$ and each $n \geq n_i$, the cohomology vanishes for every sheaf, i.e., there is an exact sequence of global sections

$$0 \longrightarrow \Gamma(X, \operatorname{im} f_{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^i(n)) \longrightarrow \Gamma(X, \operatorname{im} f_i(n)) \longrightarrow 0$$

$$0 \longrightarrow \Gamma(X, \ker f_{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i-1}(n)) \longrightarrow \Gamma(X, \operatorname{im} f_{i-1}(n)) \longrightarrow 0$$

$$0 \longrightarrow \Gamma(X, \operatorname{im} f_i(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i+1}(n)) \longrightarrow \Gamma(X, \operatorname{coker} f_i(n)) \longrightarrow 0$$

which implies the sequence

$$\Gamma(X, \mathcal{F}^{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^i(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i+1}(n))$$

is exact. Take $n_0 = \max n_i$ to obtain the desired integer. □